

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – Technical Presentation

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO1 – Imitation (L1)	Assessment:	Assessment Tool 2 – Technical Presentation
Weightage:	30% of CIA		
CO1 Statement:	Analyse NLP models, regular expression patterns, finite state automata, morphological processing techniques and to interpret language processing. (BL-3)		
Student Name:	K.KEERTHANA SAI	Reg. No.:	192425143
Section / Batch:	SLOT D	Date:	25-07-2026

Code of Conduct

I K.Keerthana sai(192425143) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: K.Keerthana Sai

Instructions

- Suggested Team Member Roles (3 Members)
- Member 1: Theory, Concepts, and Literature Review
- Member 2: Algorithm Design, Models, and Technical Analysis
- Member 3: Demonstration, Case Study, Applications, and Conclusion
- GPS photo must submit.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
Evolution of Natural Language Processing: From Rule-Based Systems to Modern NLP	NLP Models, Language Processing Evolution	TP-01
Comparative Study of NLP Models and Algorithms	Model Analysis, Algorithm Comparison	TP-02
Regular Expressions for Real-World Text Processing	Pattern Matching, Information Extraction	TP-03
Designing Efficient Regular Expression Patterns for Information Extraction	Text Mining, Data Extraction Techniques	TP-04
Finite State Automata in Natural Language Processing	State Transitions, Language Recognition	TP-05
Building a Finite State Automaton for Token Recognition	Lexical Analysis, Token Validation	TP-06
Morphology and Word Formation in Natural Languages	Morpheme Analysis, Word Structure Processing	TP-07
Inflectional Morphology: Understanding Grammatical Variations	Grammatical Analysis, Word Normalization	TP-08
Derivational Morphology and Vocabulary Expansion	Word Generation, Semantic Transformation	TP-09
Comparative Analysis of Inflectional and Derivational Morphology	Morphological Classification, Language Analysis	TP-10

Finite-State Morphological Parsing Techniques	Morphological Parsing, Finite-State Processing	TP-11
Morphological Analysis for Indian Languages	Language-Specific Morphology, NLP Challenges	TP-12
Porter Stemmer: Design, Working Mechanism, and Applications	Stemming Algorithms, Text Preprocessing	TP-13
Stemming vs Morphological Parsing: A Comparative Study	Word Reduction, Morphological Processing	TP-14
Integrated NLP Pipeline: From Regular Expressions to Morphological Processing	End-to-End Language Processing Pipeline	TP-15

Annexure A – Technical Presentation

Team No.	Technical Presentation Title	Description
1	Evolution of Natural Language Processing: From Rule-Based Systems to Modern NLP	Analyze the historical development of NLP models, key milestones, and how language processing has evolved from handcrafted rules to intelligent systems.
2	Comparative Study of NLP Models and Algorithms	Explore different NLP models and algorithms used for language processing, comparing their working principles, strengths, and limitations.
3	Regular Expressions for Real-World Text Processing	Demonstrate how regular expressions are used for data validation, text extraction, log analysis, and information retrieval in practical applications.
4	Designing Efficient Regular Expression Patterns for Information Extraction	Analyze complex regular expression patterns used to extract emails, phone numbers, URLs, dates, and structured information from unstructured text.
5	Finite State Automata in Natural Language Processing	Explain the concepts of deterministic and non-deterministic finite automata and their role in lexical analysis and language recognition.
6	Building a Finite State Automaton for Token Recognition	Design and demonstrate a finite state automaton that recognizes tokens such as identifiers, keywords, and numbers in textual data.
7	Morphology and Word Formation in Natural Languages	Investigate how words are structured and formed using morphemes, and analyze the importance of morphology in NLP systems.
8	Inflectional Morphology: Understanding Grammatical Variations	Examine how inflectional changes affect word forms based on tense, number, gender, and case across different languages.
9	Derivational Morphology and Vocabulary Expansion	Analyze derivational processes that create new words and study their impact on semantic interpretation and language understanding.
10	Comparative Analysis of Inflectional and Derivational Morphology	Compare the characteristics, applications, and challenges of inflectional and derivational morphology in NLP tasks.
11	Finite-State Morphological Parsing Techniques	Explore finite-state transducers and parsing techniques used to analyze and generate morphological structures in natural languages.
12	Morphological Analysis for Indian Languages	Study the challenges and solutions involved in performing morphological processing for morphologically rich Indian languages such as Tamil, Hindi, and Telugu.
13	Porter Stemmer: Design, Working Mechanism, and Applications	Analyze the Porter Stemming algorithm, its rule-based approach, implementation steps, and role in information retrieval systems.
14	Stemming vs Morphological Parsing: A Comparative Study	Compare stemming and morphological parsing techniques based on accuracy, computational complexity, and application scenarios.
15	Integrated NLP Pipeline: From Regular Expressions to Morphological Processing	Demonstrate a complete NLP preprocessing pipeline incorporating regular expressions, finite state automata, stemming, and morphological analysis for text processing.

Rubrics for Calculation

Technical Presentation					
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Team No. / Criteria Level	Technical Content Accuracy	Problem Identification	Use of Tools	Communication	Professionalism	Total %
3	30	25	20	12	7	97

Faculty In-charge	Course Coordinator	Program Director
Signature: T.KUMARAGURUBARGAN _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____



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DSA0308-Natural Language Processing

Integrated NLP Pipeline

From Regular Expressions to Morphological Processing

A complete NLP preprocessing pipeline using Regular Expressions,
Finite State Automata, Stemming, and Morphological Analysis

Guided by:

Dr. KUMARAGURUBARAN T

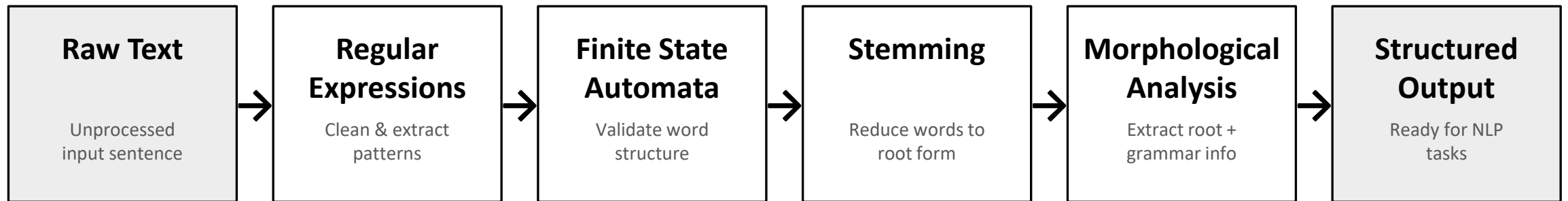
Presented by:

192425143 - K.KEERTHANA SAI

Objective

- Text data is unstructured — computers need it broken into a clean, standard form before analysis.
- This project builds one continuous pipeline that takes raw text through four processing stages.
- Each stage cleans or analyzes the text a little more, preparing it for tasks like search, translation, or sentiment analysis.
- Goal: show how simple techniques combine to solve a real NLP preprocessing problem.

The Pipeline — Big Picture



Four stages sit between raw text and structured output. Each one is explained on its own slide next.

Input	"The children were playing happily in the gardens."
Output	children→child(N,PL) playing→play(V,GER) gardens→garden(N,PL)

Step 1 — Regular Expressions

The first cleanup stage: find and handle patterns in raw text.

- A regular expression (regex) is a sequence of characters that defines a search pattern.
- Used to remove punctuation, extra spaces, numbers, or special symbols.
- Also used to extract useful patterns — emails, dates, hashtags, word boundaries.
- Fast, rule-based, and the simplest way to prepare raw text for the next stages.

Example

Input: "Hi!! Call me at 9876543210."

Pattern: `[^a-zA-Z\s]`

Action: remove non-letters

Output: "Hi Call me at "

Limitation: regex only matches surface patterns — it does not understand grammar or word structure.

Step 2 — Finite State Automata

Checking whether a word's structure is actually valid.

- A Finite State Automaton (FSA) is a machine with a fixed set of states and rules for moving between them.
- Used here to validate word structure — e.g. checking a word follows valid prefix + root + suffix patterns.
- Reads the word letter by letter (or morpheme by morpheme) and moves through states.
- If it ends in an "accept state", the word structure is valid; otherwise it is rejected.

Example — validating "unhappiness"

State q_0 → 'un' → State q_1

State q_1 → 'happi' → State q_2

State q_2 → 'ness' → Accept

Result: valid word structure ✓

FSAs are the backbone of finite-state morphological parsing used in real NLP systems.

Step 3 — Stemming (Porter Stemmer)

Cutting words down to a common root form.

- Stemming removes suffixes to reduce a word to its stem — a rough root form.
- The Porter Stemmer applies a fixed sequence of rewrite rules in phases (e.g. "-ing", "-ed", "-ational" → "-ate").
- It is fast and rule-based, but purely mechanical — it does not check real dictionary meaning.
- Widely used in search engines and information retrieval to match word variants.

Example

playing → play

happiness → happi

studies → studi

connection → connect

Note: stems like "happi" and "studi" aren't real words — that trade-off is fixed by morphological analysis.

Step 4 — Morphological Analysis

Understanding a word's real root and its grammar.

- Morphological analysis breaks a word into meaningful units called morphemes (root + prefix/suffix).
- Unlike stemming, it returns the correct dictionary root (lemma) and grammatical information.
- Especially important for morphologically rich languages such as Tamil, Hindi, and Telugu, where one root can take many suffix forms.
- Often implemented using finite-state transducers built on the FSA concept from Step 2.

Example

unhappiness → un + happy + ness

(prefix) (root) (suffix)

running → run + ing (V, GER)

gardens → garden + s (N, PL)

This is the most accurate but most computationally expensive stage in the pipeline.

Stemming vs Morphological Analysis

Aspect	Stemming	Morphological Analysis
Output	Rough stem (may not be a real word)	Correct root word (lemma)
Method	Fixed suffix-stripping rules	Grammar + FSA / FST based rules
Speed	Very fast	Slower, more processing
Accuracy	Lower	Higher
Grammar info	None	Tense, number, part of speech, etc.
Best for	Search engines, information retrieval	Translation, grammar checking, rich languages

Full Pipeline — Worked Example

Input sentence: "The runners were happily jumping over unhappiness."

1. Regex cleanup

Lowercase, remove punctuation → "the runners were happily jumping over unhappiness"

2. FSA validation

Each word checked against valid prefix–root–suffix patterns → all pass

3. Stemming

runners→runner→runn... | happily→happili | jumping→jump | unhappiness→unhappi

4. Morphological analysis

runners→run(N,PL) | happily→happy(ADV) | jumping→jump(V,GER) |
unhappiness→un+happy+ness(N)

Final structured output feeds directly into tasks like search, POS tagging, or translation.

Applications

Search Engines

Stemming/lemmatization lets a search for "run" also match "running", "runs".

Machine Translation

Morphological analysis extracts grammar so meaning transfers correctly across languages.

Spell & Grammar Checkers

FSA-based validation flags words with invalid structure.

Indian Language Processing

Morphologically rich languages like Tamil, Hindi, Telugu depend heavily on this pipeline.

Text Mining & Chatbots

Cleaned, normalized text improves keyword extraction and intent detection.

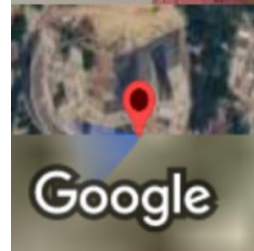
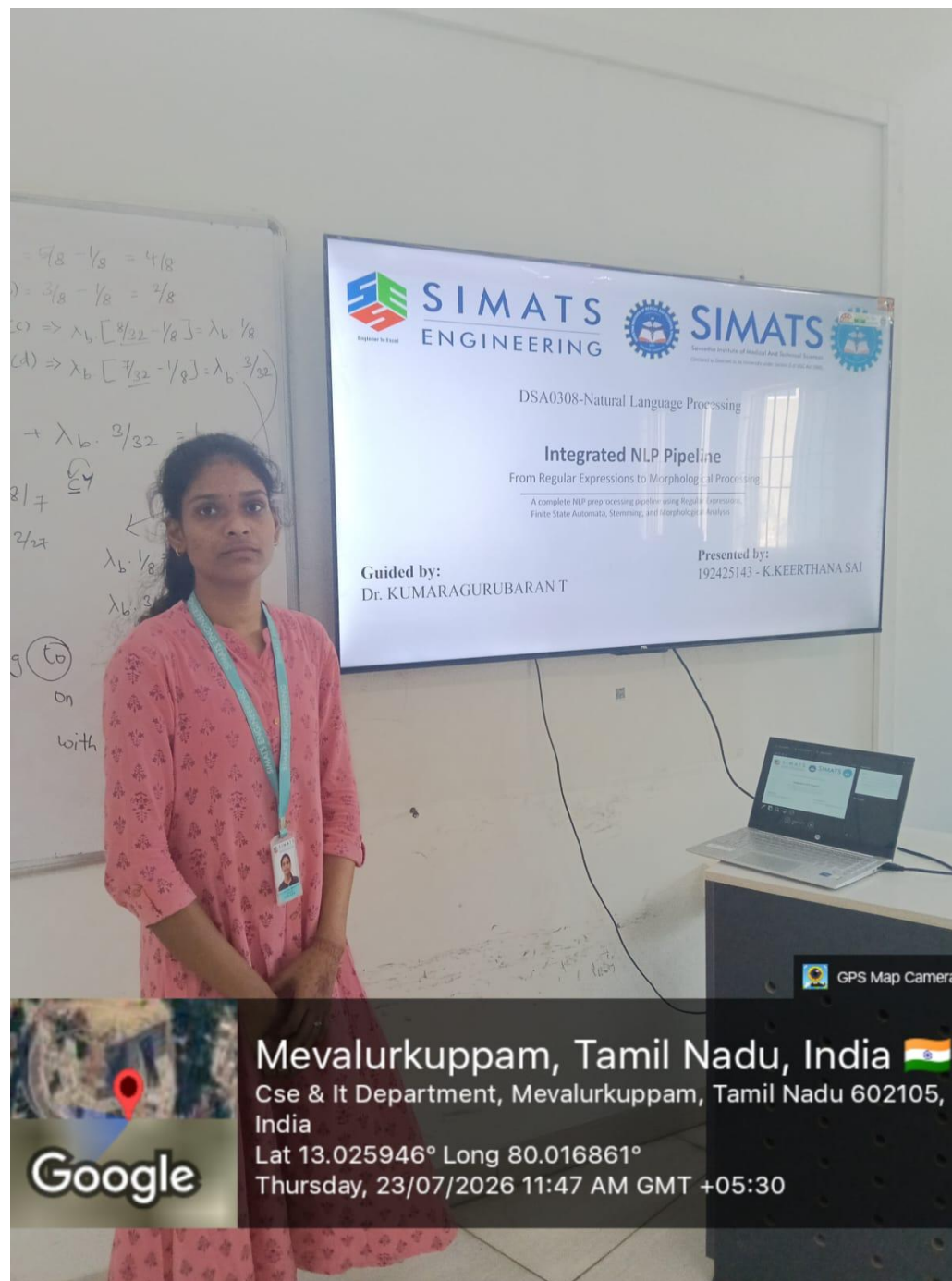
Information Retrieval

Reduces vocabulary size by grouping word variants under one root.

Conclusion

- Raw text becomes usable data only after passing through structured preprocessing.
- Regular expressions clean text; finite state automata validate word structure.
- Stemming gives a fast, rough root; morphological analysis gives an accurate root with grammar.
- Chaining all four stages builds a complete, practical NLP preprocessing pipeline — the foundation for search, translation, and text mining systems.

Thank You



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